

What is it ?

- Given 2 balls, and the N floor building, find the exact first floor that the ball will break when dropped from that floor ●_●-
- If both balls and you do not give the exact answer, that's bad ending $\varepsilon(\text{D} \cup \text{D})^3$ -
- Let's rewrite in programming term:
 - Input: An array like [0, 0, 0,...,0, 1, 1, 1, 1, 1] with size of N
 - Output: First index of element '1' when the ball break

2 search approach

- Linear search:
 - Safe solution, go from floor 1 to N step by step, we will find the answer just by 1 ball, how about the other one ? Dont forget to use it sir (;'Д'')
 - $O(N)$
- Binary search:
 - Maybe it's a better way. Or not ($= \Phi \pm \Phi =$)
 - Go to the middle floor every step
 - if the ball is broken, go lower.
 - if the ball still ok, go higher
 - but we only have 2 balls, so whenever 1 ball is broken, we still need to go search linear from last safe position
 - worst case still $O(N)$ when drop all constant, nothing change

Better way

- We can go search by jump $\sqrt{n}$ step from the beginning
- If we break 1 ball, go linear search from last safe point but we only go search at most $\sqrt{n}$ floor for the worst case
- $O(\sqrt{n})$ better than linear (■_■)

Why choose $\sqrt{n}$?

- It reduces the time complexity from linear to sub-linear, then root is the better step (especially square root is the largest among all the root) available when some other math function like logarithm can't be use or much smaller than the root

- Maybe larger step is better for this problem, because we can find first break point faster.